

Assignment Advanced Visuals in PowerBI

Q1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

- **Basic Filter:** It displays a checklist of all available values in a column where users can manually check or uncheck boxes to filter data.
- **Advanced Filter:** It allows users to apply specific text or numeric conditions like 'Contains', 'Starts with', or 'Is greater than' to isolate data.
- **Slicer:** It is an on-canvas visual filter button that lets report viewers easily click and filter charts directly from the report screen.

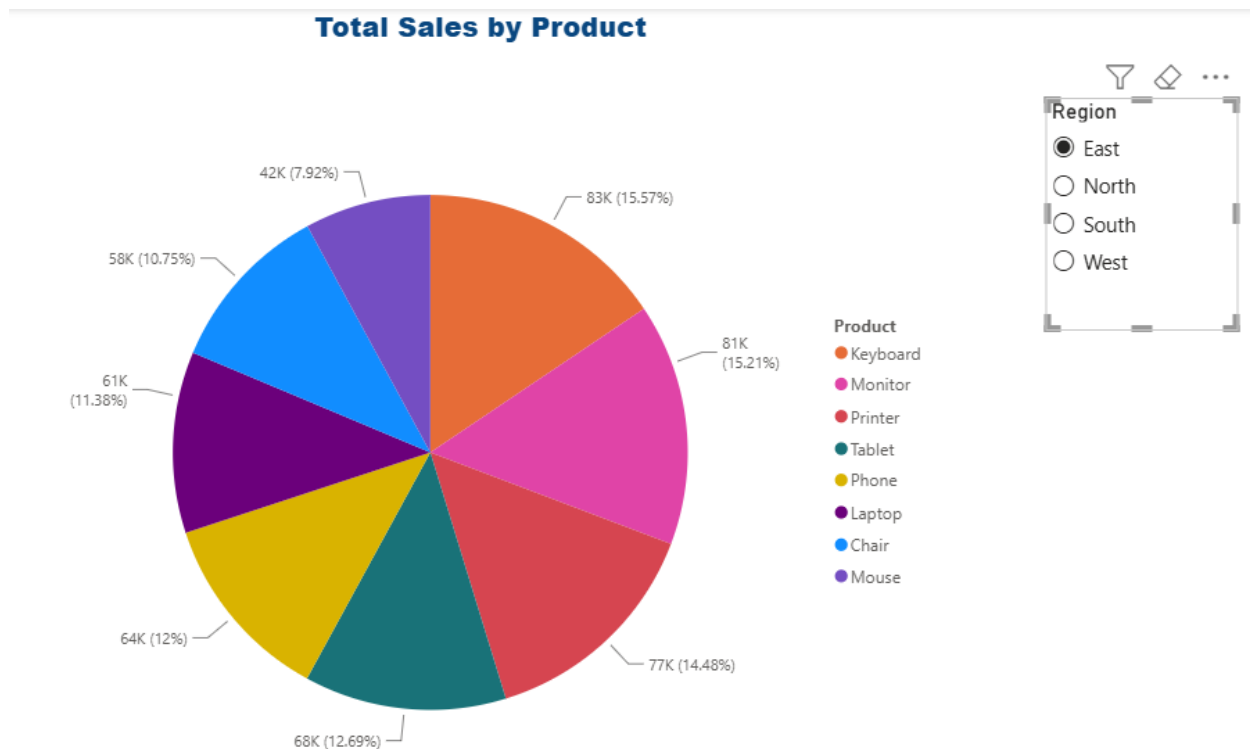
Q2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

Answer:-

The screenshot shows the Microsoft Power BI Desktop interface. The main area displays a table with the following columns: Year, Month, Customer_Name, Product, Category, Sum of Sales, Country, Region, and Sum of Quantity. The data is filtered for the year 2023. The right-hand pane shows the 'Filters' section with a search bar and two filter boxes, both containing 'Add data fields here'. The 'Visualizations' pane is also visible, showing various chart options.

Year	Month	Customer_Name	Product	Category	Sum of Sales	Country	Region	Sum of Quantity
2023	January	Aarav	Laptop	Electronics	1200	India	North	1
2023	January	Customer_102	Monitor	Accessories	7900	India	East	9
2023	January	Customer_179	Keyboard	Accessories	4030	India	North	2
2023	January	Customer_19	Laptop	Electronics	6392	India	South	7
2023	January	Customer_36	Laptop	Electronics	14440	India	South	5
2023	February	Bhavya	Phone	Electronics	800	India	South	2
2023	February	Customer_101	Mouse	Accessories	1198	India	East	1
2023	February	Customer_106	Printer	Office	5229	India	North	3
2023	February	Customer_201	Keyboard	Accessories	19914	India	North	1
2023	February	Customer_34	Printer	Office	5235	India	North	3
2023	February	Customer_94	Phone	Electronics	8043	India	West	9
2023	March	Charan	Tablet	Electronics	500	India	East	1
2023	March	Customer_134	Monitor	Accessories	5476	India	West	2
2023	March	Customer_162	Tablet	Electronics	19623	India	South	2
2023	March	Customer_217	Tablet	Electronics	16880	India	West	6
2023	March	Customer_33	Desk	Furniture	18175	India	South	7
2023	March	Customer_83	Monitor	Accessories	11651	India	South	5
2023	April	Customer_114	Monitor	Accessories	13616	India	East	9
2023	April	Customer_133	Laptop	Electronics	6497	India	North	3
2023	April	Customer_157	Printer	Office	10767	India	South	7
2023	April	Customer_188	Laptop	Electronics	9946	India	East	4
2023	April	Customer_205	Phone	Electronics	4100	India	West	3
2023	April	Customer_224	Tablet	Electronics	11882	India	West	4
2023	April	Customer_44	Chair	Furniture	14757	India	East	3
2023	April	Customer_5	Laptop	Electronics	6896	India	West	9
2023	April	Customer_6	Laptop	Electronics	19442	India	South	3
2023	April	Devansh	Monitor	Accessories	300	India	West	3
2023	May	Customer_113	Mouse	Accessories	1044	India	South	4
Total					726999			143

Q3. Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region.



Q4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

- **Visual-Level Filter:** It applies to only one specific chart on the canvas and does not affect any other charts.
- **Page-Level Filter:** It applies to all charts and visuals present on that particular report page.
- **Report-Level Filter:** It applies across the entire Power BI report, filtering data on all pages simultaneously.

Q5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.

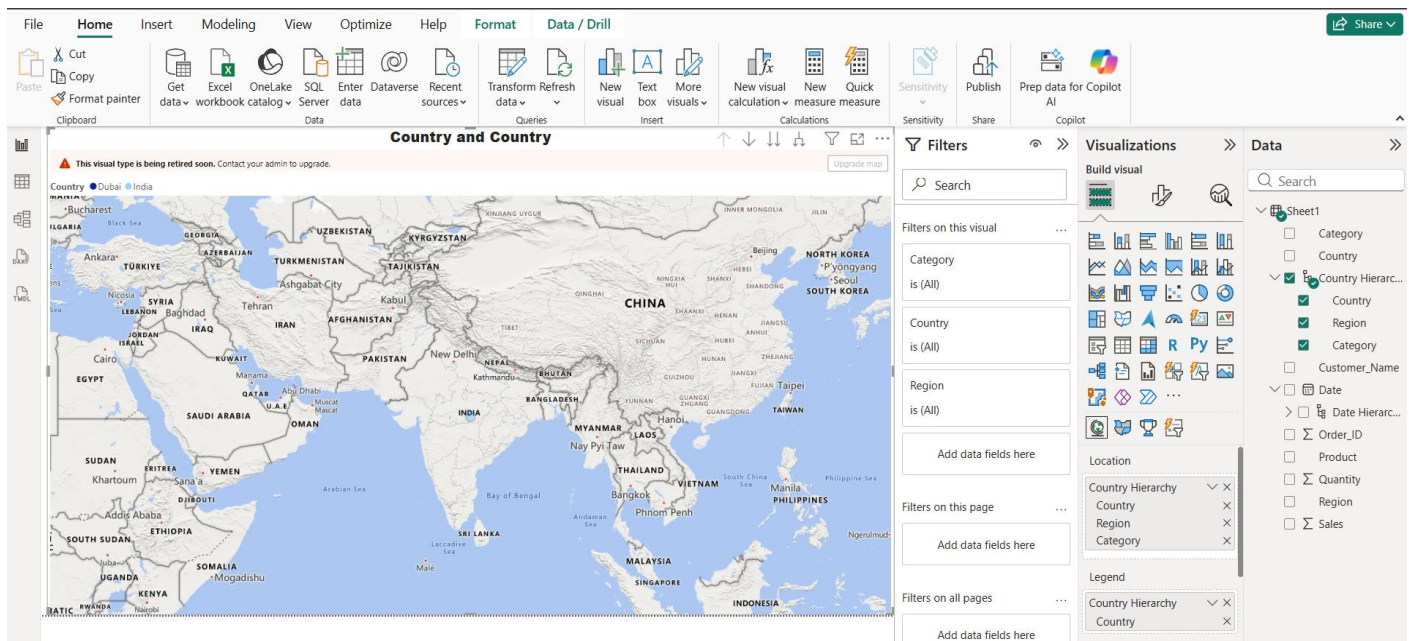
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2023	January	Customer_102	Monitor	Accessories	7900	India	East	9
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2023	February	Customer_106	Printer	Office	5229	India	North	3
2023	February	Customer_201	Keyboard	Accessories	19914	India	North	1
2023	February	Customer_54	Printer	Office	5235	India	North	3
2023	February	Customer_94	Phone	Electronics	8043	India	West	9
2023	March	Charan	Tablet	Electronics	500	India	East	1
2023	March	Customer_134	Monitor	Accessories	5476	India	West	2
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2023	April	Customer_6	Laptop	Electronics	19442	India	South	3
2023	April	Devansh	Monitor	Accessories	300	India	West	3
2023	May	Customer_111	Mouse	Accessories	7004	India	North	4
Total					726899			338

Q6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

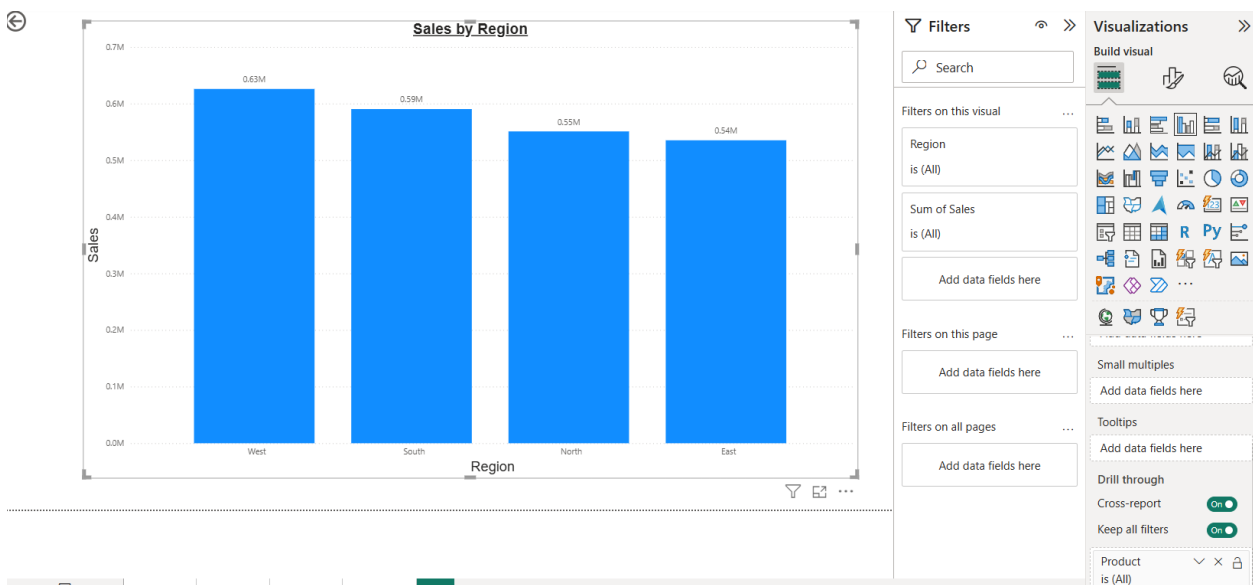
- **Role:** Drill-Down lets users navigate to a deeper, more detailed level inside a predefined data hierarchy (like moving from Year to Month).
- **Improvement:** It improves data exploration by allowing users to start with a high-level summary view and click into a specific category to see its underlying breakdown without changing the page.

Q7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.

Hierarchy_Structure = Right click on Country in data field and create Hierarchy -> Right click on Region and add to Country Hierarchy -> Right click on Category and add to Country Hierarchy

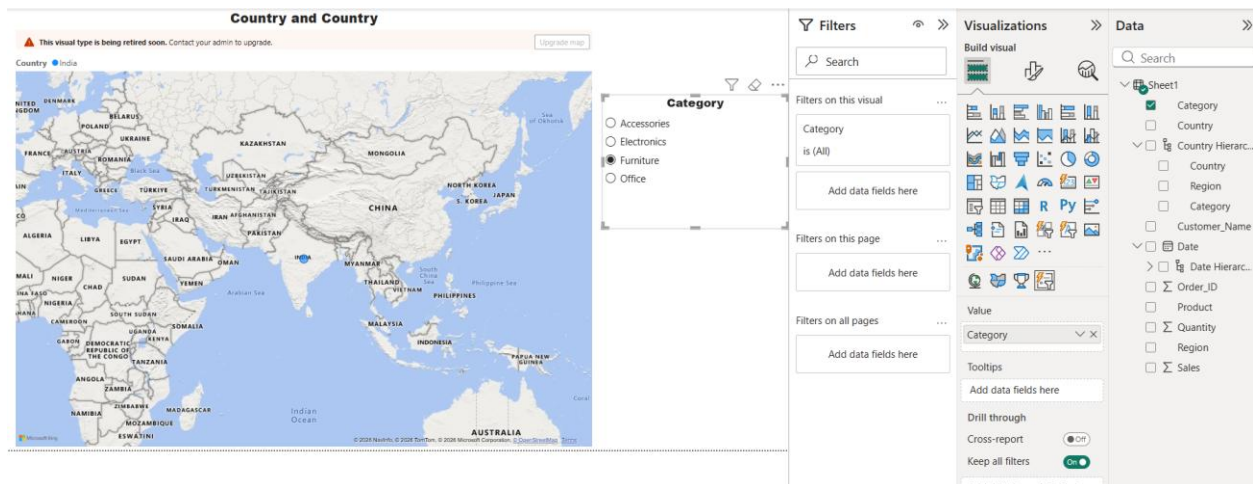


Q8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region-wise sales



Q9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

Hierarchy view with slicer:-



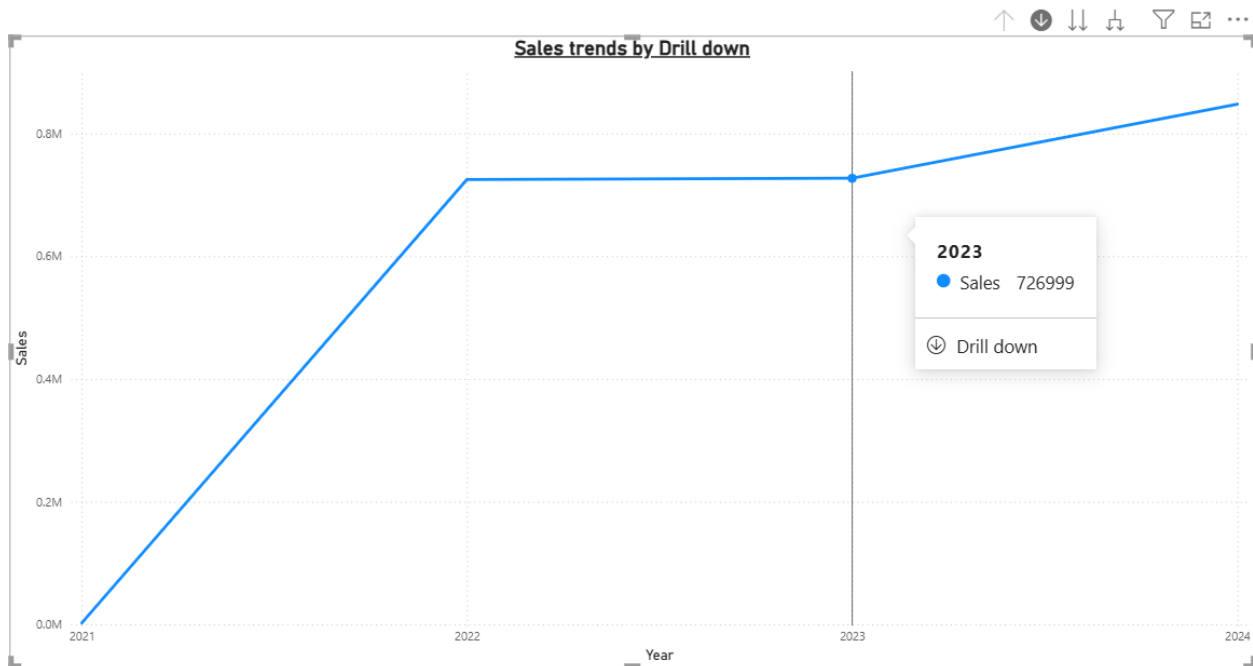
Interaction with the Drill-Down Hierarchy: -

When you select a Product Category in the slicer, the hierarchy chart will automatically filter to show only that category's data.

Q10. Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

- **Drill-Down Example:** In a time-series sales chart, clicking on the year **2023** expands the same visual to reveal months (Jan, Feb, Mar) inside that same chart without navigating away.
- **Drill-Through Example:** In a high-level product table, right-clicking **"Laptop"** opens a completely separate, detailed report page that displays inventory levels and supplier metrics specifically for laptops.

Q11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.



Q12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000. Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.

DAX: Total Sales = SUM(Sales_Dataset[Sales])

Visual Output (Sales > 50,000):

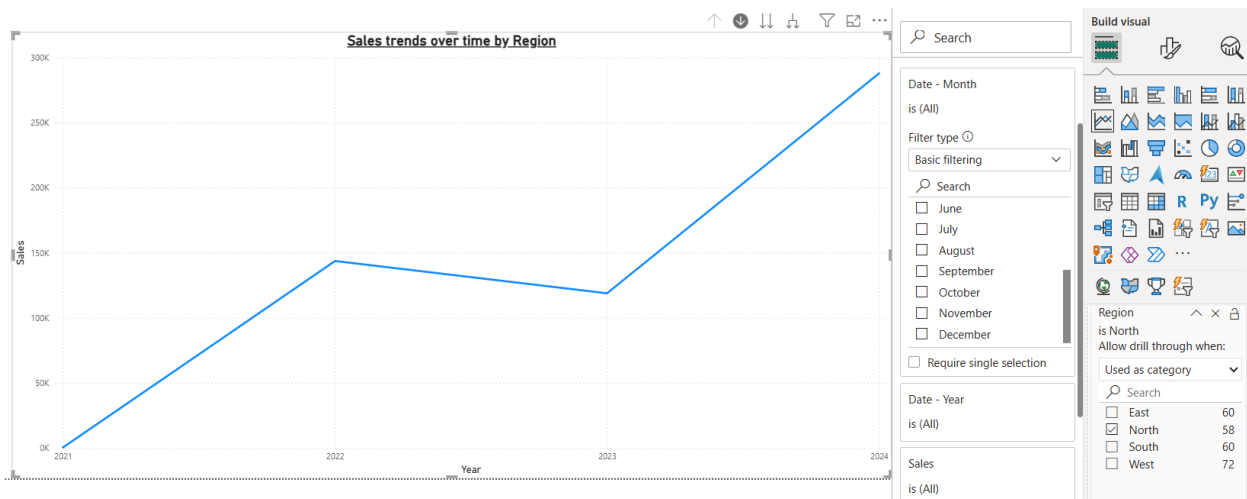
Product	Total Sales	Filter_Status
Keyboard	₹74,387	Displayed
Laptop	₹54,611	Displayed
Monitor	₹23,280	Hidden

Q13. What is the "Keep All Filters" option in Drill-Through? How does it affect user interaction?

- **Definition:** It is a configuration toggle located in the Drill-through settings pane of a destination page.

- **Effect:** When switched **On**, any slicers or filters active on the source page (like a selected year or region) transfer to the destination page. When **Off**, only the specific right-clicked data point filters the target page.

Q14. Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.



Q15. Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data.

Dashboard_Layout:

Page_1: Overview (Slicers, Pie Chart, Line Chart)

Page_2: Product Detail (Drill-Through Target)

